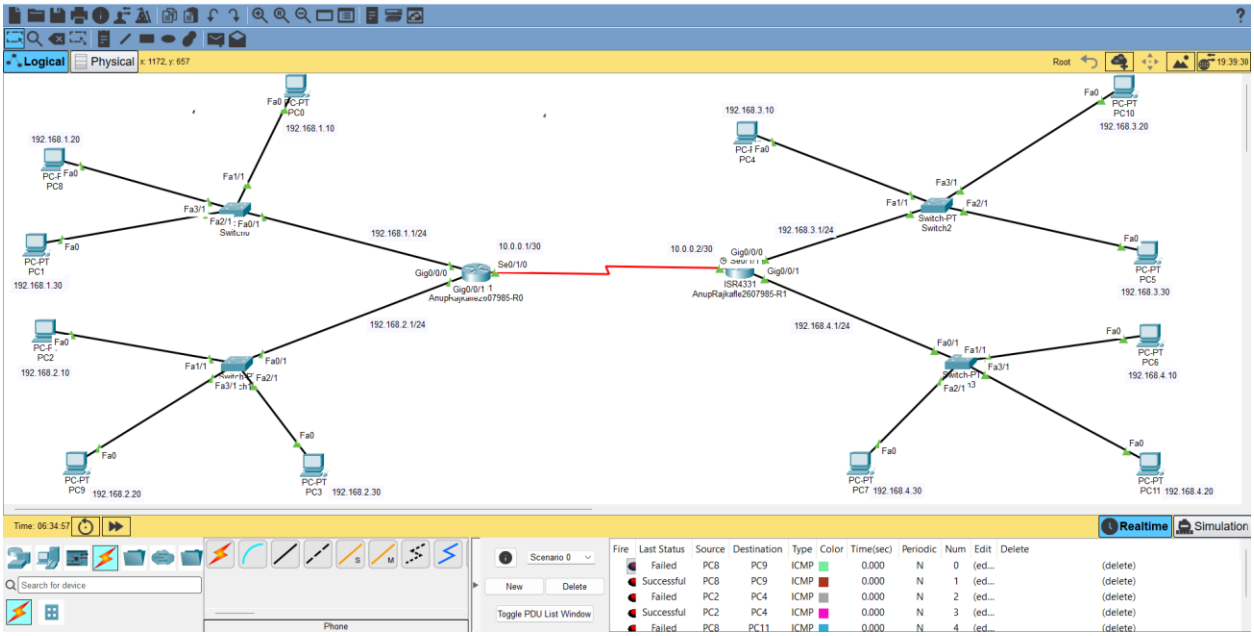
Network Topology Screenshot



Router Configuration Logs

CLI commands from export log (hostname, interface setup, static routes).

```
Router>

Router>en

Router#config t

Router(config)#hostname AnupRajkafle2607985-R1

AnupRajkafle2607985-R1(config)#interface GigabitEthernet0/0/0

AnupRajkafle2607985-R1(config-if)#ip address 192.168.3.1 255.255.255.0

AnupRajkafle2607985-R1(config-if)#no shutdown

AnupRajkafle2607985-R1(config)#interface GigabitEthernet0/0/1

AnupRajkafle2607985-R1(config-if)#ip address 192.168.4.1 255.255.255.0

AnupRajkafle2607985-R1(config-if)#no shutdown

AnupRajkafle2607985-R1(config)#interface Serial0/1/1

AnupRajkafle2607985-R1(config-if)#ip address 10.0.0.2 255.255.255.252

AnupRajkafle2607985-R1(config-if)#no shutdown

AnupRajkafle2607985-R1(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1

AnupRajkafle2607985-R1(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.1
```

Mapping topology

LAN	Switch	Router Interface	PCs (from your labels)
LAN1	Switch0	Router0 Gig0/0/0	PC0, PC1, PC4
LAN2	Switch1	Router0 Gig0/0/1	PC2, PC3, PC9
LAN3	Switch2	Router1 Gig0/0/0	PC5, PC8, PC10
LAN4	Switch3	Router1 Gig0/0/1	PC6, PC7, PC11
WAN		Router0 Se0/1/0 ↔ Router1 Se0/1/1	

IP Addressing Plan

Segment	Network	Router IP
LAN1	192.168.1.0/24	192.168.1.1 (R0 Gig0/0/0)
LAN2	192.168.2.0/24	192.168.2.1 (R0 Gig0/0/1)
LAN3	192.168.3.0/24	192.168.3.1 (R1 Gig0/0/0)
LAN4	192.168.4.0/24	192.168.4.1 (R1 Gig0/0/1)
WAN	10.0.0.0/30	R0 Se0/1/0: 10.0.0.1 / R1 Se0/1/1: 10.0.0.2

PC addressing:

PC	IP	Mask	Gateway
PC0	192.168.1.10	255.255.255.0	192.168.1.1
PC1	192.168.1.11	255.255.255.0	192.168.1.1
PC4	192.168.1.12	255.255.255.0	192.168.1.1
PC2	192.168.2.10	255.255.255.0	192.168.2.1
PC3	192.168.2.11	255.255.255.0	192.168.2.1
PC9	192.168.2.12	255.255.255.0	192.168.2.1
PC5	192.168.3.10	255.255.255.0	192.168.3.1
PC8	192.168.3.11	255.255.255.0	192.168.3.1
PC10	192.168.3.12	255.255.255.0	192.168.3.1
PC6	192.168.4.10	255.255.255.0	192.168.4.1
PC7	192.168.4.11	255.255.255.0	192.168.4.1
PC11	192.168.4.12	255.255.255.0	192.168.4.1

Verification Evidence Screenshots

Use the following commands for validation and include their screenshots in your report: show Ip interface brief, show ip route, ping traceroute.

Verify successful connectivity between PCs in different LANs.

AnupRajkafle2607985-R0

```
AnupRajkafle2607985-R0>en
AnupRajkafle2607985-R0#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   192.168.1.1     YES manual up                    up
GigabitEthernet0/0/1   192.168.2.1     YES manual up                    up
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
Serial0/1/0            10.0.0.1        YES manual up                    up
Serial0/1/1            unassigned      YES unset  administratively down down
AnupRajkafle2607985-R0#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

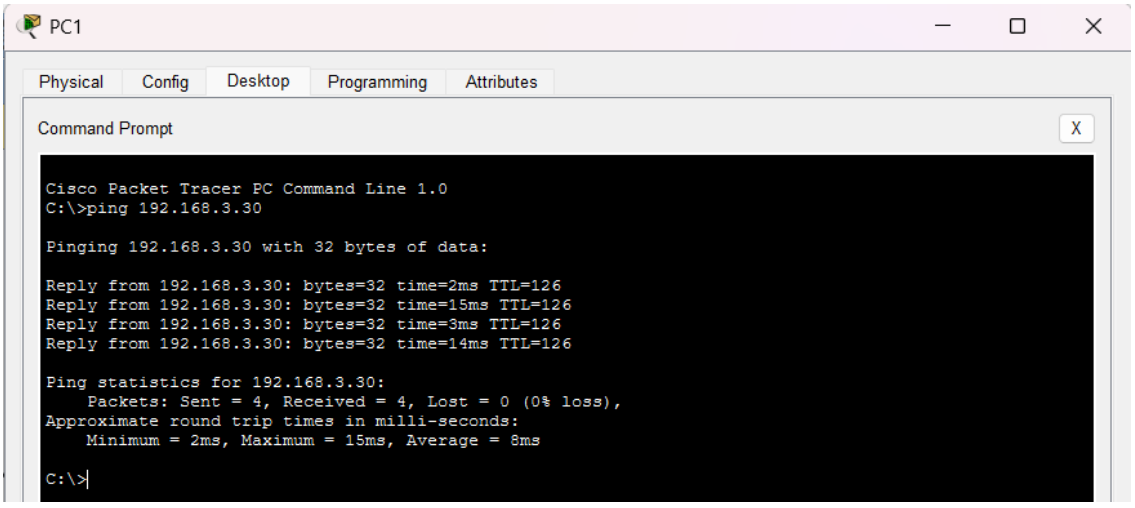
     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/1/0
L       10.0.0.1/32 is directly connected, Serial0/1/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0/0
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.2.1/32 is directly connected, GigabitEthernet0/0/1
S    192.168.3.0/24 [1/0] via 10.0.0.2
S    192.168.4.0/24 [1/0] via 10.0.0.2

AnupRajkafle2607985-R0#ping 192.168.3.30

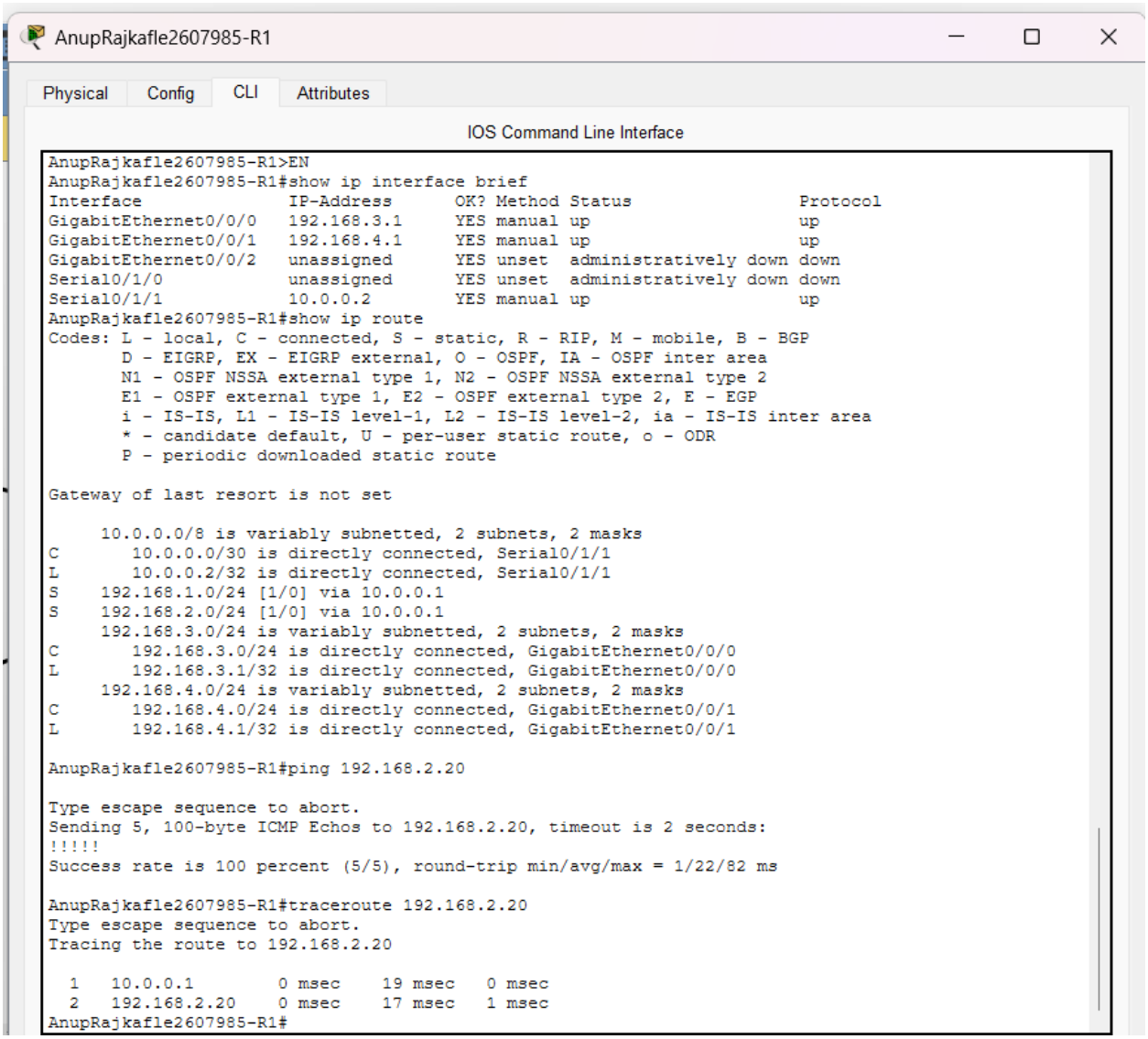
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.30, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/6/20 ms

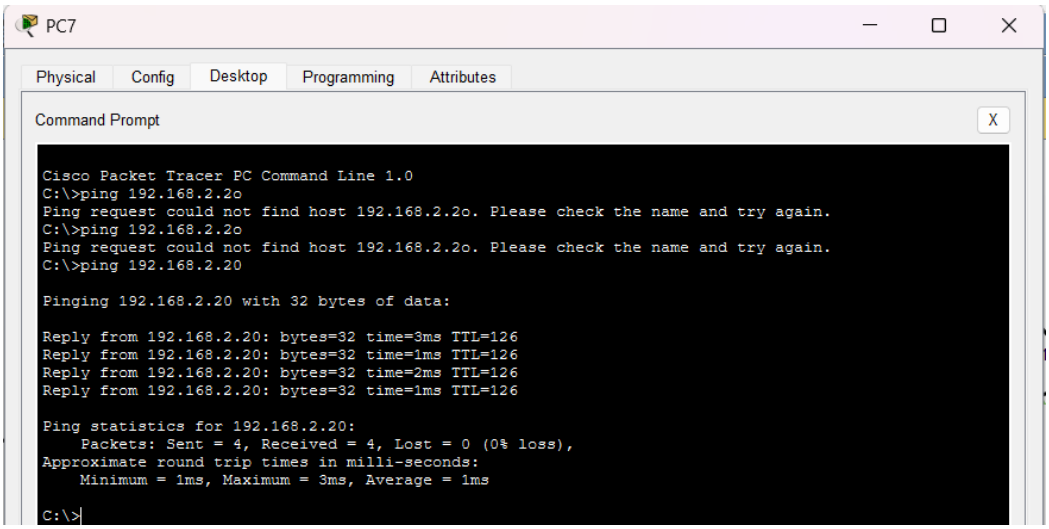
AnupRajkafle2607985-R0#traceroute 192.168.3.30
Type escape sequence to abort.
Tracing the route to 192.168.3.30

  1   10.0.0.2        15 msec    8 msec    15 msec
  2   192.168.3.30    0 msec     0 msec    8 msec
AnupRajkafle2607985-R0#
```



AnupRajkafle2607985-R1





Brief Summary

Static routing is a method that allows routers to forward packets to other networks without any direct connection by specifying manually the next hop path to each remote network. In the current laboratory, all static routes of the routers will be pointing to the WAN side of the other router in order to allow the PCs from two separate networks, such as LAN1 and LAN3, to communicate through both routers. Otherwise, they would not be able to connect to each other at all.